New Evidence on Deliberative Distortions: An Out-of-Sample Replication and Extension

Working draft, September 2026

Abstract

Do small-group discussions produce similar patterns of homogenization, polarization, and domination outside the samples in which those patterns were originally studied? We apply the measures of Luskin et al. (2022) to 13 previously unused studies from 12 public datasets. The sample includes 3 Deliberative Polls and 10 studies of other discussion formats. The main comparison uses paired policy responses and equal weights across study families. Mean homogenization is 1.42, 1.23, and 0.76 points on a 0–100 scale in the original DPs, new DPs and other group discussions, respectively. Mean absolute midpoint-distance changes are positive in all three sets. Weighting and omission of individual families materially affect some results; only two dependence families supply the new DP comparison. Whole-group movement toward advantaged participants' initial positions has no consistent sign across the three sets and demographic dimensions. The education decomposition is more consistent: in each study with those data, disadvantaged members move on average toward the advantaged reference, while advantaged members move in the opposite direction. Small whole-group averages can therefore coexist with appreciable subgroup and individual movement. These comparisons describe the available study samples; they do not identify causal effects of discussion formats or the mechanisms of influence.

1 The question

Small average changes in the distribution of opinions can conceal considerable movement by individual discussants. They can also conceal different experiences across discussions. Luskin et al. (2022) distinguished three features of attitude change: declining dispersion within a group, movement of its mean away from a scale midpoint, and movement toward the initial position of an advantaged subgroup. Their evidence came from Deliberative Polls. We examine whether the same descriptive patterns recur in unused polls and whether they characterize other discussion designs.

This comparison has two parts. The replication applies the original measures to new Deliberative Poll datasets, with an available-response analysis matching the original sampling convention. The extension applies those measures to other observed discussions, including town meetings, classroom and laboratory groups, partisan groups, and dyadic conversations. We emphasize paired respondents in the extension so that changes in group composition between surveys do not generate apparent attitude movement. We report the same paired benchmark for the corrected original sample, as well as the available-response comparison.

The current evidence does not support a single description of polarization across settings. A1R 2019 and A1R Climate have positive directional polarization, whereas Tanzania has negative directional polarization. Other discussion formats also vary. Moreover, movement against a group's

initial direction can carry it across the midpoint and farther from neutrality on the opposite side. The original directional measure and a change in absolute midpoint distance can consequently have opposite signs. Both are reported, with their distinct denominators.

Domination requires a second distinction. The whole group's movement need not represent the movement of either subgroup. Our parallel to the original paper's Table 3 preserves the direction and magnitude of movement separately for advantaged and disadvantaged members. In the education comparisons, both components recur more consistently than the sign of their weighted aggregate. This is evidence about the geometry of attitude change. It does not establish which participants persuaded others, or whether any change resulted from weighing reasons, social pressure, shared information, or another process.

2 Measures and comparisons

Let A_{igjt} be respondent i's attitude on item j in discussion group g at wave t, rescaled to $[0, 1]$ using the instrument's endpoints. Let $\bar{A}_{gjt}$ and s_{gjt} be the group mean and sample standard deviation. The DP replication uses available responses at each wave, matching the original. The extension retains the same observed respondents at both waves for each item. A respondent can therefore contribute to one item but not another.

Following Luskin et al. (2022), homogenization is

$$H_{gj} = s_{gj1} - s_{gj2}. \tag{1}$$

Positive values indicate lower dispersion within the discussion group. Directional polarization is

$$P_{gj} = (\bar{A}_{gj2} - \bar{A}_{gj1}) \operatorname{sgn}(\bar{A}_{gj1} - m_j), \tag{2}$$

where m_j is the instrument-supported neutral midpoint. We leave P undefined when the initial mean equals the midpoint or the midpoint's meaning is unverified. The separate absolute-distance measure is

$$P^{\text{abs}}_{gj} = |\bar{A}_{gj2} - m_j| - |\bar{A}_{gj1} - m_j|. \tag{3}$$

It remains defined at an initial midpoint. Thus differences between the two summaries can reflect both midpoint crossings and different eligible cells.

For domination, let $\bar{A}^a_{gj1}$ be the advantaged subgroup's initial mean. Define $q_{gj} = \operatorname{sgn}(\bar{A}^a_{gj1} - \bar{A}_{gj1})$ and

$$D_{gj} = q_{gj}(\bar{A}_{gj2} - \bar{A}_{gj1}). \tag{4}$$

A positive value means movement toward the advantaged initial reference; a negative value means movement in the opposite direction. The definitions of advantage follow the original where the instruments permit: male, above-median education, above-median income, and their intersection. Socioeconomic thresholds are computed within the participating event sample before item-specific missingness. Missing demographics are excluded from calculations requiring them.

The corresponding movements of disadvantaged and advantaged members are $M^d_{gj} = q_{gj}(\bar{A}^d_{gj2} - \bar{A}^d_{gj1})$ and $M^a_{gj} = q_{gj}(\bar{A}^a_{gj2} - \bar{A}^a_{gj1})$. With a fixed paired sample,

$$D_{gj} = d_{gj}M^d_{gj} + (1 - d_{gj})M^a_{gj}, \tag{5}$$

where d_{gj} is the disadvantaged share among the respondents contributing to that group-item calculation. The code checks this identity within every eligible cell. It does not generally hold after

replacing its cell-specific weights and components by their separate averages. Negative M^a means advantaged members move toward the disadvantaged subgroup's initial position; it is not a negative amount of individual attitude change.

Groups must have both demographic categories for a domination calculation. Unavailable subgroup means remain missing. A tied initial reference leaves the direction undefined; zero movement with a defined direction remains an observed zero. H requires at least two observed respondents per wave. A sensitivity analysis requires at least five. These rules are particularly consequential for dyads and triads, where demographic homogeneity or one missing response can remove a group from a particular calculation.

We summarize signed means and the proportion of eligible cells greater than zero, matching the original paper's magnitude and occurrence distinction. Group-item episodes receive equal weight within a reported study-condition and construct. We also report equal-family/equal-event weighting and leave-family-out summaries. Small groups, repeated items and repeated conditions do not create independent study replications. The inferential unit for cross-study summaries is the study family. CR2 covariance estimates and Satterthwaite degrees of freedom follow the existing analysis protocol; intervals are suppressed when there are fewer than two families or effective degrees of freedom below four. Occurrence frequencies use the same procedure applied to binary outcomes. The present compatible format/construct/timing strata do not support the reported protocol's pooled intervals, so the main comparisons are descriptive.

3 Data

3.1 Previously unused Deliberative Polls

The DP replication includes A1R 2019, A1R Climate and Tanzania. A1R 2019 gathered nationally recruited delegates in Dallas for discussion of multiple policy areas (Fishkin et al., 2021). A1R Climate used an online format and a different issue agenda. Returning participants link the two studies, which are assigned one dependence family. Neither delegate sample is treated as an unrestricted random sample of all adults for these unweighted group-level summaries.

The Tanzania study followed deliberation about natural resources and public policy (Sandefur et al., 2022). The February 2015 baseline (t0) preceded information sessions, and the May–July 2015 telephone follow-up (t3) occurred after the discussion event. The comparison therefore spans information, deliberation and a longer interval than an immediately administered pre/post questionnaire. Participants were reassigned between discussion rounds. We retain the 371 released household-to-group assignments across 22 policy items, including the borrowing item on its released five-category scale, and keep the repeated participants within one family.

3.2 Other discussion designs

The extension deliberately reaches beyond the DP design. Our Budget, Our Economy used local in-person discussion tables connected to a national broadcast. Its sites share one national event. Celaya's laboratory study contributes deliberation sessions; its Q&A and packet-only comparisons are outside the current observed- discussion analysis. Hong Kong supplies one facilitated and one casual discussion group on Article 23, with selected strong supporters and opponents (Shen and Yu, 2021). The facilitated condition also supplied an information booklet. Its small number of groups sharply limits inference about differences between conditions.

The UK study contributes same-party and mixed-party Zoom discussions of immigration (Hobolt et al., 2023). Separate source studies contribute US cross-party text conversations and five-day WhatsApp conversations in India (Majumdar, 2026). These latter datasets primarily concern affect or stereotypes. Their group-level polarization measures are changes relative to each item's neutral midpoint, not the customary difference between ratings of an in-party and an out-party. They are presented separately from policy attitudes.

The two consensus studies observe conversations among Dutch pupils and German university students (Koudenburg et al., 2023). Random formation of groups in the pupil study does not supply a discussion-versus-no-discussion treatment effect; random group formation is not documented for the student study. The Diplomacy experiment compares structured and unstructured face-to-face laboratory discussions in the same triads (Roos et al., 2022). Topics were selected for initial disagreement, and each triad discussed two topics from a larger topic bank. Finally, Shizuoka's facilitated mini-public addressed who should participate in decisions about a nuclear restart (Nakazawa et al., 2024). Its information-accuracy belief is separated from its decision-procedure preferences.

Table 1: Studies in the out-of-sample replication and extension

Study	Country	Year	Mode	N	Episodes	Items	Policy
A1R 2019	United States	2019	Face-to-face	523	40	47	47
Our Budget, Our Economy	United States	2010	Face-to-face	2751	339	6	6
Hong Kong	Hong Kong	2020	Face-to-face	12	2	1	1
Tanzania	Tanzania	2015	Face-to-face	371	50	22	22
A1R Climate	United States	2021	Online	962	105	72	65
Celaya	United States	2018	Face-to-face	104	7	8	8
UK partisan discussion	United Kingdom	2021	Online	653	87	4	4
WhatsApp India	India	Unknown	Online	712	356	18	0
US cross-party conversation	United States	2021	Online	294	147	9	0
Consensus: pupils	Netherlands	Unknown	Face-to-face	116	28	2	2
Consensus: students	Germany	Unknown	Face-to-face	136	41	2	2
Diplomacy lab	Netherlands	2020	Face-to-face	105	70	10	8
Shizuoka	Japan	2019	Face-to-face	65	8	11	10

N counts retained participant records within a study before item-specific pairing and demographic exclusions. Episodes count actual discussion-group episodes; Tanzania reassigns participants, and Diplomacy reuses triads across conditions. Items count the released item bank used here, not the number answered by every participant. Policy is the subset classified as policy attitudes. Dates are fieldwork dates; unknown dates are not replaced by publication years. OBOE's local tables were face-to-face although its national plenary was broadcast-linked. Source archives are listed in Table 6.

3.3 Coverage and selection

Table 1 contains 6,804 participant-study records across 13 studies and 11 dependence families. These are not counts of independent people or independent experiments. Participation and repeated measurement are required for the analysis, and some sources condition on completed conversations. The original study's events are excluded; older unused studies can qualify without being temporal replications. The original comparison uses the corrected repository outputs, not a transcription of the published tables.

This is a retrospective secondary analysis. Source papers and the initial H/P/D results had been inspected before this standalone draft and subgroup decomposition. Public-data discovery remains open. The screening register contains 101 records reviewed through archive queries, project pages and bounded citation follow-up. It is not a systematic census. Public microdata from Shin-Kori and Denmark contain repeated attitudes but lack verified discussion-group assignments. Some Finnish datasets require researcher access. Other promising projects supply aggregate reports, unpublished panelist files, or nominal choices that do not meet the current scalar measurement specification. These access and measurement barriers may select which settings enter the comparison.

The September 12 follow-up added source leads but no eligible dataset. The public six-country antimicrobial-resistance file contains 1,280 treatment respondents and 1,139 controls, with no discussion-room mapping. Malawi's 2017 flood-policy poll reports 468 post-survey respondents in 24 groups, but its microdata are available on author request. Additional A1R, Meta, African and Mongolian polls have public reports; Danish and Korean national archives identify further deposits whose access or variables require resolution. Finland's FSD2975 is openly downloadable, contrary to our initial screening record, but its released questionnaire lacks verified matched policy waves and actual discussion-group IDs. The source register documents each disposition; these leads are not counted as included studies.

4 Results

4.1 Overall comparison

The central comparison is between the original DPs, the new DPs, and the other group discussions, without selecting settings on the basis of their results. Table 2 uses paired policy responses throughout and reports means on a 0–100 attitude scale. Panel A weights each dependence family equally and, within a family, each eligible study equally. Each study mean averages its eligible group-item episodes, combining conditions rather than selecting the favorable one. The original polls each constitute one family. The three new DPs constitute two families because the two A1R studies share participants. Eight other studies contribute policy outcomes in seven families. Affective dyadic conversations have different outcomes and are reported separately, not folded into policy averages.

Equal-family weighting prevents the largest archive from dominating the comparison, but gives small studies substantial weight. Panel B therefore gives every eligible group-item episode equal weight, matching the original aggregate convention. Neither weighting estimates a representative average over all possible discussions. Outcome-specific family counts are essential: education D in the new DPs comes only from A1R Climate, and income and combined D in other discussions come from only two families. These demographic averages cannot establish an overall format advantage.

The overall means show modest homogenization: 1.42 points in the original DPs, 1.23 in the new DPs, and 0.76 in other discussions. Absolute extremity increases by 2.01, 1.91, and 2.44 points,

respectively. Thus agreement and movement toward neutrality do not coincide in the aggregate. The new-DP directional P is -1.01 with equal family weights but 1.83 with equal group-item weights. Its sign depends on how the two available families are weighted, so it cannot support a stable claim that new DPs depolarize or polarize more than other discussions. The ordering of absolute P also reverses: with group-item weights, it is 3.83 in new DPs and 2.84 in other discussions.

Whole-group gender D is 0.10, -0.16, and -1.12 points in the three sets. Education D is 1.44, 0.91, and 0.37 points. These signed means provide no uniform pattern of movement favoring advantage across discussion formats; demographic coverage differs, and small signed means can conceal larger opposing movements. The subgroup decomposition below makes that distinction explicit.

Omitting one family at a time puts other-discussion homogenization between -0.71 and 2.43 points. In particular, omitting Hong Kong's twelve-person pilot changes it to 2.43; equal family weighting gives that pilot the same weight as the much larger OBOE study. Other-discussion absolute P remains positive under every single-family omission, between 1.73 and 3.06. These ranges describe sensitivity to the observed corpus, not confidence intervals. The data support modest average changes and substantial heterogeneity, rather than a general ranking of which discussion format works best.

The two dyadic conversation studies contribute affective ratings rather than policy attitudes. Their equal-family averages are 1.60 points for H, -4.77 for directional P, and 0.60 for absolute P. These summarize changes in feelings toward rated targets; they do not measure the gap between political parties or religious groups. They extend the evidence on agreement and extremity but cannot enter the same substantive average as policy outcomes.

Table 2: Overall comparison of original DPs, new DPs and other group discussions

A. Equal weight to families, then studies

Study set	H	Directional P	Absolute P	Gender D	Education D	Income D	Combined D
Original DPs	1.42 (21)	-0.86 (21)	2.01 (21)	0.10 (21)	1.44 (21)	0.26 (11)	1.02 (11)
New DPs	1.23 (2)	-1.01 (2)	1.91 (2)	-0.16 (2)	0.91 (1)	0.50 (1)	0.60 (1)
Other group discussions	0.76 (7)	-0.36 (7)	2.44 (7)	-1.12 (6)	0.37 (3)	-0.55 (2)	-0.94 (2)

B. Equal weight to eligible group-item episodes

Study set	H	Directional P	Absolute P	Gender D	Education D	Income D	Combined D
Original DPs	1.51 (21)	-1.27 (21)	1.35 (21)	0.12 (21)	1.33 (21)	0.27 (11)	1.40 (11)
New DPs	2.86 (2)	1.83 (2)	3.83 (2)	0.19 (2)	0.91 (1)	0.50 (1)	0.60 (1)
Other group discussions	1.31 (7)	-1.18 (7)	2.84 (7)	-0.15 (6)	0.30 (3)	0.51 (2)	-0.51 (2)

Cells report means on a 0–100 scale, with the number of contributing dependence families in parentheses. All responses are paired, metric by metric. Positive H means lower dispersion; positive directional P means movement further along the group's initial midpoint direction; positive absolute P means increased distance from neutrality. Positive D means movement toward the initial advantaged reference. Original outcomes are indices and OOS outcomes are individual items. Families contribute different topics, samples and discussion designs. These are descriptive averages, not causal effects of format; intervals are omitted because the new DP comparison has only two families. The CSVs include eligible pair counts, positive frequencies, both membership rules and omission of each family.

4.2 Magnitudes and occurrence

Table 3 parallels the original paper's magnitude and occurrence table. The DP panel uses available-wave responses. For A1R 2019, H is 0.035 and directional P is 0.013. A1R Climate gives 0.030 and 0.022; Tanzania gives -0.008 and -0.037. The corrected original's available-wave benchmarks are 0.016 and -0.013. Thus the new polls do not reproduce a uniform combination of slight homogenization and movement against the initial midpoint direction.

Table 3: Homogenization, polarization and domination: magnitude and occurrence

A. Deliberative Poll replication: available-wave responses

Study / condition	H	P	Gender D	Education D	Income D	Combined D
Corrected original	0.016 / 59.2	-0.013 / 48.1	0.001 / 48.2	0.014 / 54.7	0.003 / 50.3	0.014 / 54.4
A1R 2019	0.035 / 69.9	0.013 / 59.0	0.006 / 50.5	–	–	–
Tanzania	-0.008 / 47.1	-0.037 / 37.3	-0.010 / 45.1	–	–	–
A1R Climate	0.030 / 65.3	0.022 / 61.2	0.001 / 48.8	0.010 / 52.8	0.005 / 49.7	0.006 / 50.4

B. Other designs and corrected-original benchmark: paired responses

Study / condition	H	P	Gender D	Education D	Income D	Combined D
Corrected original	0.015 / 57.0	-0.013 / 45.8	0.001 / 46.2	0.013 / 52.0	0.003 / 49.7	0.014 / 53.7
Our Budget, Our Economy	0.008 / 44.3	-0.019 / 39.3	0.005 / 43.2	0.003 / 44.6	0.006 / 44.1	-0.004 / 43.3
Hong Kong: facilitated	-0.145 / 0.0	0.000 / 0.0	–	–	–	–
Hong Kong: casual	-0.040 / 0.0	–	–	–	–	–
Celaya	-0.015 / 33.9	-0.031 / 28.3	-0.011 / 40.7	-0.019 / 32.7	-0.017 / 35.8	-0.014 / 36.7
UK: same-party	0.052 / 74.5	0.060 / 76.1	-0.019 / 43.2	–	–	–
UK: mixed-party	0.005 / 54.4	-0.013 / 40.3	-0.006 / 41.7	–	–	–
Consensus: pupils	0.024 / 55.4	–	-0.036 / 34.1	–	–	–
Consensus: students	0.013 / 28.0	-0.006 / 23.0	0.002 / 33.3	–	–	–
Diplomacy: structured	0.129 / 74.1	-0.011 / 31.8	0.015 / 41.2	–	–	–
Diplomacy: unstructured	0.064 / 57.1	0.045 / 40.9	-0.069 / 11.8	–	–	–
Shizuoka	0.008 / 48.8	-0.014 / 41.0	-0.005 / 40.4	0.027 / 52.0	–	–

Each cell reports signed mean / percentage positive. Panel A uses available-wave responses; Panel B uses paired respondents. Both give equal weights for eligible group-item episodes within the study-condition. H, P and D are on the 0–1 attitude scale; 0.01 is one percentage point of the full scale. The corrected original uses policy indices, whereas the new studies use individual items. A dash indicates no eligible estimate, not a zero. Counts differ by measure and are provided in the accompanying estimates CSV. Conditions within a study are not independent studies. These are descriptive estimates; supported pooled intervals are unavailable under the prespecified family/df rule.

The distributional comparison retains the original paper's zero-centered displays (Figure 1). Each new poll has its own row. Pooling the rows would allow the study with the largest number of groups and items to determine much of the appearance. Figure 2 extends the comparison to other discussion conditions. Both figures show heterogeneity rather than an estimated causal contrast between discussion designs.

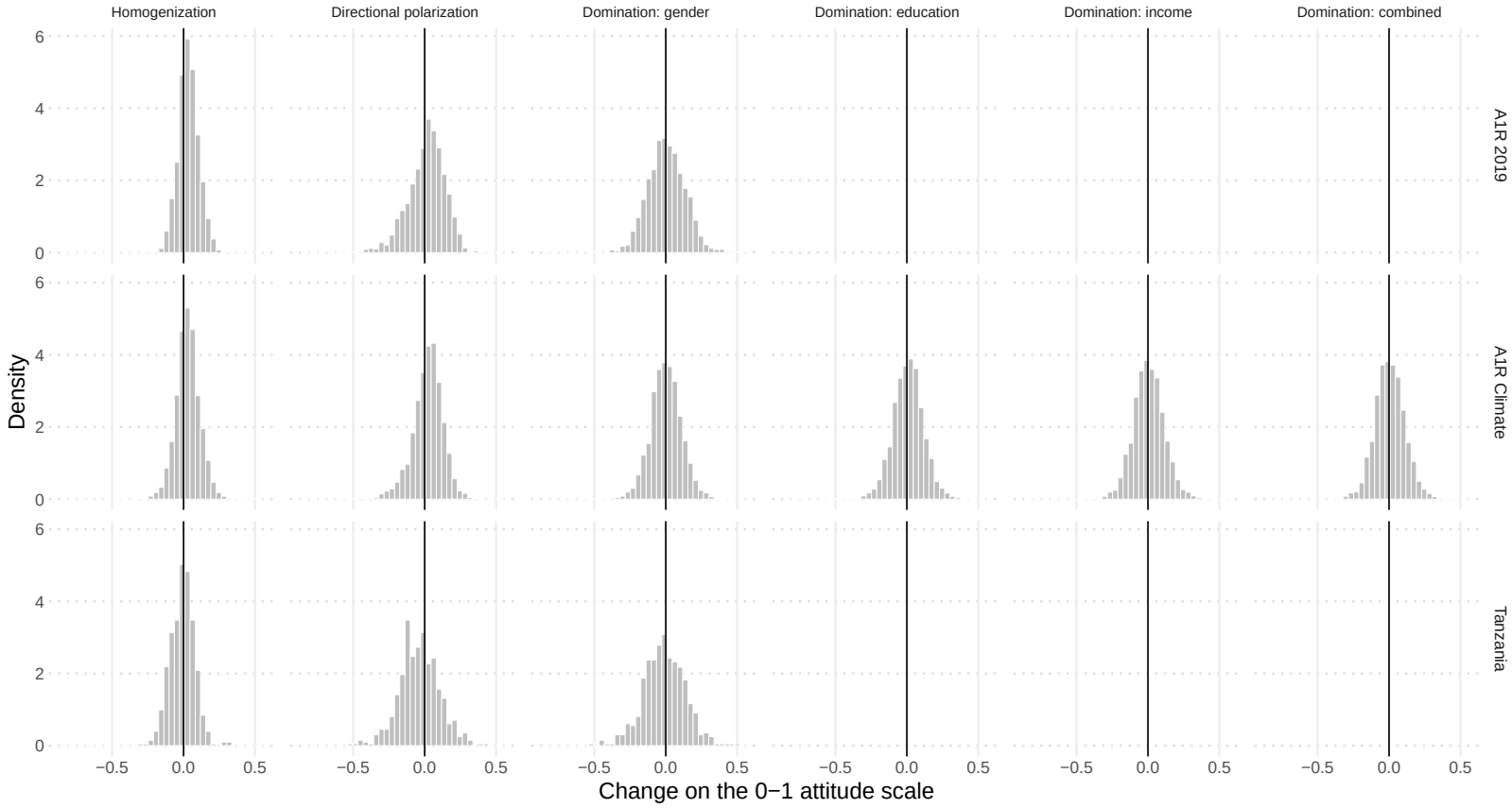


Figure 1: Distributions of available-wave H, P and D in the new Deliberative Polls. Bars are histograms of eligible group-item episodes, normalized to density within each panel; vertical lines mark zero. Axes are shared. Empty panels indicate no eligible demographic comparison. Items and groups within a poll are dependent; these distributions are descriptive, not sampling distributions.

The distinction between polarization definitions changes some apparent conclusions. In Shizuoka, directional P is -0.014, while absolute-distance P is 0.009. OBOE similarly gives -0.019 and 0.028. A negative directional mean does not by itself establish movement closer to neutrality. Appendix B reports both definitions and their eligible counts.

4.3 Parsing domination

The subgroup comparisons use paired responses throughout, so that composition is fixed and the weighted identity holds. Whole-group D varies across settings. The original's positive education benchmark, 0.013, is 0.009 in A1R Climate, 0.003 in OBOE, -0.019 in Celaya and 0.027 in Shizuoka. These contrasts concern movement toward initial subgroup positions. They are not direct measurements of power or persuasion.

Table 4 separates the components. In the corrected original, the disadvantaged education subgroup's mean movement is 0.038, compared with -0.037 for advantaged members in the same signed direction. In A1R Climate the corresponding means are 0.033 and -0.064. The disadvantaged component is positive and the advantaged component negative in each OOS study with education data. Their directions therefore recur even though the sign of whole-group D does not. These are signed movements toward initial positions; midpoint or subgroup-reference crossings can complicate descriptions of the final distance between subgroups.

4.4 Attitude change

Small H, P or D need not mean respondents barely changed their views. We parallel the original paper's net/gross comparison using complete respondent pairs. Net change is the absolute mean individual change within a group-item episode; gross change is the mean absolute individual change. Opposing individual movements can cancel in the former. Table 5 averages each quantity across eligible group-item episodes. The companion CSV also averages episodes equally after aggregating their items, matching the original's alternative weighting.

5 What replicates, and what remains open

The evidence separates three questions that are easy to combine. First, do unused Deliberative Polls reproduce the original signed averages? Not uniformly: the new polls differ in directional polarization and one differs in homogenization. Second, do other discussion formats show the same averages? Their descriptive results also vary, with substantial differences in who participates, which topics are discussed, group size, timing and measurement. A contrast between two rows cannot isolate the effect of facilitation, composition, mode or any other design feature.

Third, does the distinction between whole-group and subgroup movement matter outside the original study? The education decomposition indicates that it does. A whole-group mean close to zero can accompany larger subgroup movements in opposing signed directions. Reporting only D would omit that pattern. It would also leave unanswered whether limited net movement merely reflects limited individual change, which is why the gross-change comparison is part of the parallel analysis.

The present study is an analysis of observed pre/post discussions, including selected treatment or conversation-completer samples. Some source experiments contain randomized contrasts that are

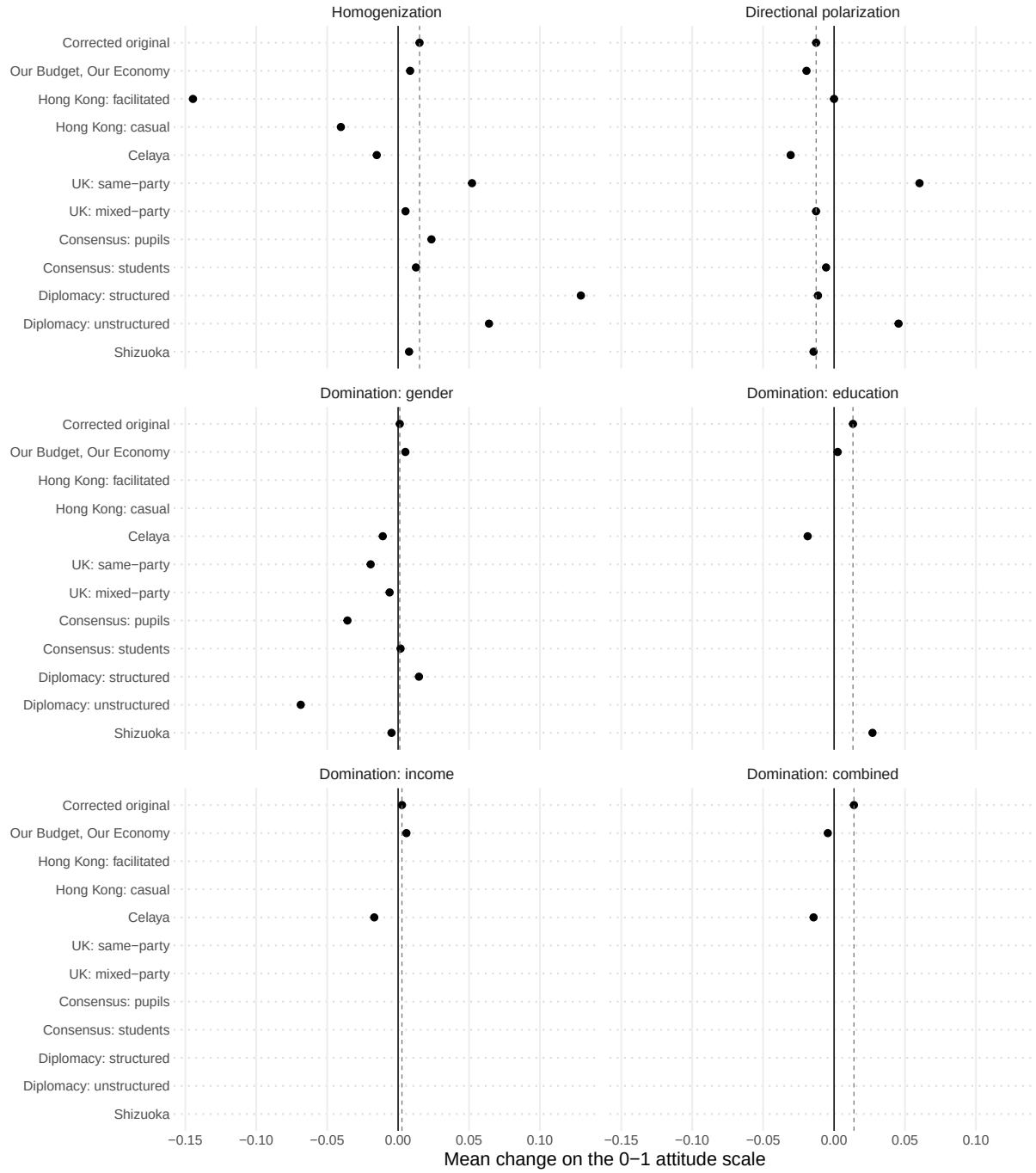


Figure 2: Paired policy estimates in the corrected original and other discussion designs. Points are signed means across eligible group-item episodes; dashed lines are the corrected original's paired benchmark for each measure. Conditions and study labels identify the observed settings, not randomized cross-study treatment contrasts. Points have no inferential intervals because these are descriptive study summaries. The original measures indices and the extension measures individual items.

Table 4: Parsing domination: whole-group and subgroup movement

Gender

Study / condition	Whole group D	Disadvantaged M	Advantaged M
Corrected original	0.001 / 46.2 (2396)	0.040 / 54.7 (2396)	-0.037 / 35.4 (2396)
A1R 2019	0.007 / 50.2 (1853)	0.051 / 57.4 (1853)	-0.035 / 38.3 (1853)
Our Budget, Our Economy	0.005 / 43.2 (969)	0.068 / 46.2 (969)	-0.052 / 25.6 (969)
Tanzania	-0.007 / 45.8 (541)	0.072 / 65.1 (541)	-0.093 / 30.7 (541)
A1R Climate	0.001 / 47.5 (6502)	0.046 / 55.5 (6502)	-0.048 / 32.5 (6502)
Celaya	-0.011 / 40.7 (54)	0.034 / 51.9 (54)	-0.087 / 22.2 (54)
UK: same-party	-0.019 / 43.2 (162)	0.013 / 42.6 (162)	-0.054 / 32.1 (162)
UK: mixed-party	-0.006 / 41.7 (156)	0.027 / 50.0 (156)	-0.039 / 30.1 (156)
Consensus: pupils	-0.036 / 34.1 (41)	0.025 / 34.1 (41)	-0.124 / 24.4 (41)
Consensus: students	0.002 / 33.3 (24)	-0.010 / 29.2 (24)	-0.026 / 20.8 (24)
Diplomacy: structured	0.015 / 41.2 (17)	0.118 / 47.1 (17)	-0.074 / 11.8 (17)
Diplomacy: unstructured	-0.069 / 11.8 (17)	-0.029 / 23.5 (17)	-0.125 / 11.8 (17)
Shizuoka	-0.005 / 40.4 (57)	0.055 / 42.1 (57)	-0.064 / 24.6 (57)

Education

Study / condition	Whole group D	Disadvantaged M	Advantaged M
Corrected original	0.013 / 52.0 (2373)	0.038 / 56.9 (2373)	-0.037 / 37.0 (2373)
Our Budget, Our Economy	0.003 / 44.6 (1493)	0.047 / 45.9 (1493)	-0.052 / 25.6 (1493)
A1R Climate	0.009 / 51.5 (5913)	0.033 / 57.2 (5913)	-0.064 / 26.9 (5913)
Celaya	-0.019 / 32.7 (52)	0.012 / 48.1 (52)	-0.110 / 13.5 (52)
Shizuoka	0.027 / 52.0 (75)	0.087 / 60.0 (75)	-0.049 / 29.3 (75)

Income

Study / condition	Whole group D	Disadvantaged M	Advantaged M
Corrected original	0.003 / 49.7 (1144)	0.037 / 54.4 (1144)	-0.046 / 38.5 (1144)
Our Budget, Our Economy	0.006 / 44.1 (1486)	0.060 / 45.1 (1486)	-0.054 / 26.1 (1486)
A1R Climate	0.005 / 48.8 (6379)	0.041 / 55.7 (6379)	-0.044 / 33.8 (6379)
Celaya	-0.017 / 35.8 (53)	0.024 / 56.6 (53)	-0.073 / 15.1 (53)

Combined advantage

Study / condition	Whole group D	Disadvantaged M	Advantaged M
Corrected original	0.014 / 53.7 (810)	0.026 / 56.2 (810)	-0.060 / 29.6 (810)
Our Budget, Our Economy	-0.004 / 43.3 (397)	0.021 / 45.6 (397)	-0.066 / 17.4 (397)
A1R Climate	0.006 / 49.1 (3659)	0.024 / 54.6 (3659)	-0.084 / 23.5 (3659)
Celaya	-0.014 / 36.7 (30)	-0.004 / 40.0 (30)	-0.108 / 16.7 (30)

Cells report mean / percentage positive (eligible group-item episodes). All samples are paired; each component retains its own defined direction and available means. Positive disadvantaged M means movement toward the advantaged initial mean. Negative advantaged M means movement toward the disadvantaged initial mean. The within-cell weighted decomposition is exact; separately averaged components need not reproduce average D. All estimates are descriptive.

not estimated here. Such within-study contrasts could address narrower causal questions, provided their assignment, attrition, outcome and dependence structures are carried through explicitly. They would answer a different question from pooling unrelated formats into a single regression.

The remaining data work is concrete. Several public datasets need discussion-group linkage; others require access to participant files or a verified baseline instrument. The current search does not justify a claim about the universe of all deliberative studies. Expanding the usable corpus and harmonizing original indices with new items could change both the comparison and its precision.

Table 5: Absolute net and gross attitude change on policy items

Study	Net	Gross	Pairs
Corrected original	0.087	0.200	2476
A1R 2019	0.102	0.200	1880
Our Budget, Our Economy	0.116	0.186	1986
Hong Kong: facilitated	0.000	0.133	1
Hong Kong: casual	0.033	0.067	1
Tanzania	0.116	0.352	550
A1R Climate	0.085	0.170	6825
Celaya	0.054	0.256	56
UK: same-party	0.087	0.131	188
UK: mixed-party	0.064	0.125	160
Consensus: pupils	0.115	0.148	56
Consensus: students	0.060	0.080	82
Diplomacy: structured	0.127	0.157	27
Diplomacy: unstructured	0.104	0.158	28
Shizuoka	0.106	0.198	80

All quantities use paired respondents on the 0–1 instrument scale. Net is the absolute mean respondent change within a group-item episode; gross is the mean absolute respondent change. Both columns then weight group-item episodes equally. The original row uses its policy indices. Pairs counts episodes with at least one complete respondent. These descriptive summaries do not identify why people changed.

References

James Fishkin, Alice Siu, Larry Diamond, and Norman Bradburn. Is Deliberation an Antidote to Extreme Partisan Polarization? Reflections on "America in One Room". *American Political Science Review*, 2021. doi: 10.1017/S0003055421000642. URL https://doi.org/10.1017/S0003055421000642.

Sara B. Hobolt, Katharina Lawall, and James Tilley. The Polarizing Effect of Partisan Echo Chambers. *American Political Science Review*, 2023. doi: 10.1017/S0003055423001211. URL https://doi.org/10.1017/S0003055423001211.

Namkje Koudenburg, Maja Kutlaca, and Toon Kuppens. The experience and emergence of attitudinal consensus in conversations. *European Journal of Social Psychology*, 2023. doi: 10.1002/ejsp.2992. URL https://doi.org/10.1002/ejsp.2992.

Robert C. Luskin, Gaurav Sood, James S. Fishkin, and Kyu S. Hahn. Deliberative Distortions? Homogenization, Polarization, and Domination in Small Group Discussions. *British Journal of Political Science*, 2022. doi: 10.1017/S0007123421000168. URL https://doi.org/10.1017/S0007123421000168.

Rajeshwari Majumdar. Reducing Prejudice and Support for Religious Nationalism Through Conversations on WhatsApp. *Comparative Political Studies*, 2026. doi: 10.1177/00104140261418627. URL https://doi.org/10.1177/00104140261418627.

Takashi Nakazawa, Tomoyuki Tatsumi, Yume Souma, and Susumu Ohnuma. An Effect of Storytelling on Attitude Changes in Deliberative Mini-Publics. *Journal of Deliberative Democracy*, 2024. doi: 10.16997/jdd.1426. URL https://doi.org/10.16997/jdd.1426.

Carla Anne Roos, Tom Postmes, and Namkje Koudenburg. Attempts to encourage diplomacy in online interactions: Three informative failures. *Acta Psychologica*, 2022. doi: 10.1016/j.actpsy.2022.103661. URL https://doi.org/10.1016/j.actpsy.2022.103661.

Justin Sandefur, Nancy Birdsall, James Fishkin, and Mujobu Moyo. Democratic Deliberation and the Resource Curse. *World Politics*, 2022. doi: 10.1017/S0043887122000090. URL https://doi.org/10.1017/S0043887122000090.

Fei Shen and Wenting Yu. Reducing political polarization in Hong Kong: a pilot experiment of deliberation. *Japanese Journal of Political Science*, 2021. doi: 10.1017/S1468109921000335. URL https://doi.org/10.1017/S1468109921000335.

A Sources, instruments and eligibility

Table 6: Public archives used in this analysis

Study	Archive	Released
A1R 2019	10.7910/DVN/KJ8IH2	2021-07-27
Our Budget, Our Economy	10.7910/DVN/XDDN2Y	2019-09-10
Hong Kong	10.7910/DVN/W9DAQM	2021-10-26
Tanzania	10.7910/DVN/S3NRQL	2022-09-27
A1R Climate	10.7910/DVN/IIOG1S	2024-11-25
Celaya	10.7910/DVN/OOMED8	2019-05-15
UK partisan discussion	10.7910/DVN/BM2A1Q	2023-12-01
WhatsApp India	10.7910/DVN/WYXEXN	2026-01-04
US cross-party conversation	10.7910/DVN/EVRMBG	2023-11-14
Consensus: pupils	10.34894/6IOPCF	2023-12-13
Consensus: students	10.34894/6IOPCF	2023-12-13
Diplomacy lab	10.34894/SAZIAE	2022-04-14
Shizuoka	10.6084/m9.figshare.22229794	2023-03-07

The accompanying `items.csv` is the item-level appendix: it records the released labels, pre/post columns, instrument endpoints, missing codes, neutral midpoint, construct and episode. It should be read with the linked source instruments; not every archive label is verbatim questionnaire wording. `events.csv` records study, condition, country, mode, date precision, post timing and dependence family. `source_register.csv` records the specific evidence and unresolved requirements for included and deferred studies. Source downloads are checked against the SHA-256 hashes in `files.csv`.

Harmonization does not convert rankings or nominal task choices into cardinal attitudes. The pupil consensus study retains its fractional seven-point responses, but supplies no P because a neutral midpoint label is unverified. Shizuoka's procedural preferences and information belief remain separate. Climate policy, concern and belief items are separate, as are affect and stereotypes in the conversation studies. No public table number is substituted for missing respondent microdata or discussion assignments.

The original's conceptual Figures 1–2 illustrate definitions and theoretical expectations. They do not have new empirical observations to reproduce here. Equations in this manuscript supply the definitions; the empirical distribution figures parallel the original's Figure 3. Its Tables 1–3 correspond to our inventory, magnitude/occurrence table and subgroup decomposition.

B Sampling and measurement sensitivity

Table 7: Full available-wave comparison and extension sampling sensitivity

Study / condition	H	P	Gender D	Education D	Income D	Combined D
Corrected original	0.016 / 59.2	-0.013 / 48.1	0.001 / 48.2	0.014 / 54.7	0.003 / 50.3	0.014 / 54.4
A1R 2019	0.035 / 69.9	0.013 / 59.0	0.006 / 50.5	–	–	–
Our Budget, Our Economy	0.017 / 51.1	-0.017 / 46.5	0.008 / 49.1	0.005 / 50.1	0.010 / 49.3	0.012 / 50.4
Hong Kong: facilitated	-0.145 / 0.0	0.000 / 0.0	–	–	–	–
Hong Kong: casual	-0.040 / 0.0	–	–	–	–	–
Tanzania	-0.008 / 47.1	-0.037 / 37.3	-0.010 / 45.1	–	–	–
A1R Climate	0.030 / 65.3	0.022 / 61.2	0.001 / 48.8	0.010 / 52.8	0.005 / 49.7	0.006 / 50.4
Celaya	-0.015 / 33.9	-0.031 / 28.3	-0.011 / 40.7	-0.019 / 32.7	-0.017 / 35.8	-0.014 / 36.7
UK: same-party	0.050 / 74.5	0.059 / 75.0	-0.022 / 44.8	–	–	–
UK: mixed-party	0.004 / 53.1	-0.016 / 38.9	-0.009 / 40.4	–	–	–
Consensus: pupils	0.024 / 55.4	–	-0.036 / 34.1	–	–	–
Consensus: students	0.013 / 28.0	-0.006 / 23.0	0.002 / 33.3	–	–	–
Diplomacy: structured	0.129 / 74.1	-0.011 / 31.8	0.015 / 41.2	–	–	–
Diplomacy: unstructured	0.064 / 57.1	0.045 / 40.9	-0.069 / 11.8	–	–	–
Shizuoka	0.007 / 50.0	-0.016 / 41.0	-0.006 / 40.4	0.027 / 52.0	–	–

Cells report mean / percentage positive using all observed respondents at each wave. The DP and original entries reproduce Panel A of Table 3; the extension entries are a sampling sensitivity relative to its paired Panel B. Differences can reflect changing respondent composition. Scale and missing-value conventions are unchanged.

The minimum-five restriction, event/family weights and leave-family-out results are regenerated in `tabs/`. The restriction intentionally removes dyads and triads from H/P comparisons; it is not a robustness test on an unchanged target population. The WhatsApp source flags possible duplicates or associates, rather than establishing duplicate identities. Its primary sample retains them, and a separate sensitivity excludes flagged respondents using the same score definitions.

The correlation counterpart to the original's text analysis is `paper_correlations.csv`, computed separately by study-condition and construct with pairwise eligible cells. `paper_composition.csv` parallels its regressions on disadvantaged share using paired policy samples. Those fits are descriptive within-study associations. Predictions at shares 0.2 and 0.8 are retained only when inside the observed share range. They do not extrapolate across unrelated discussion formats, and no independent-pair standard errors are reported.

C Distributions across discussion designs

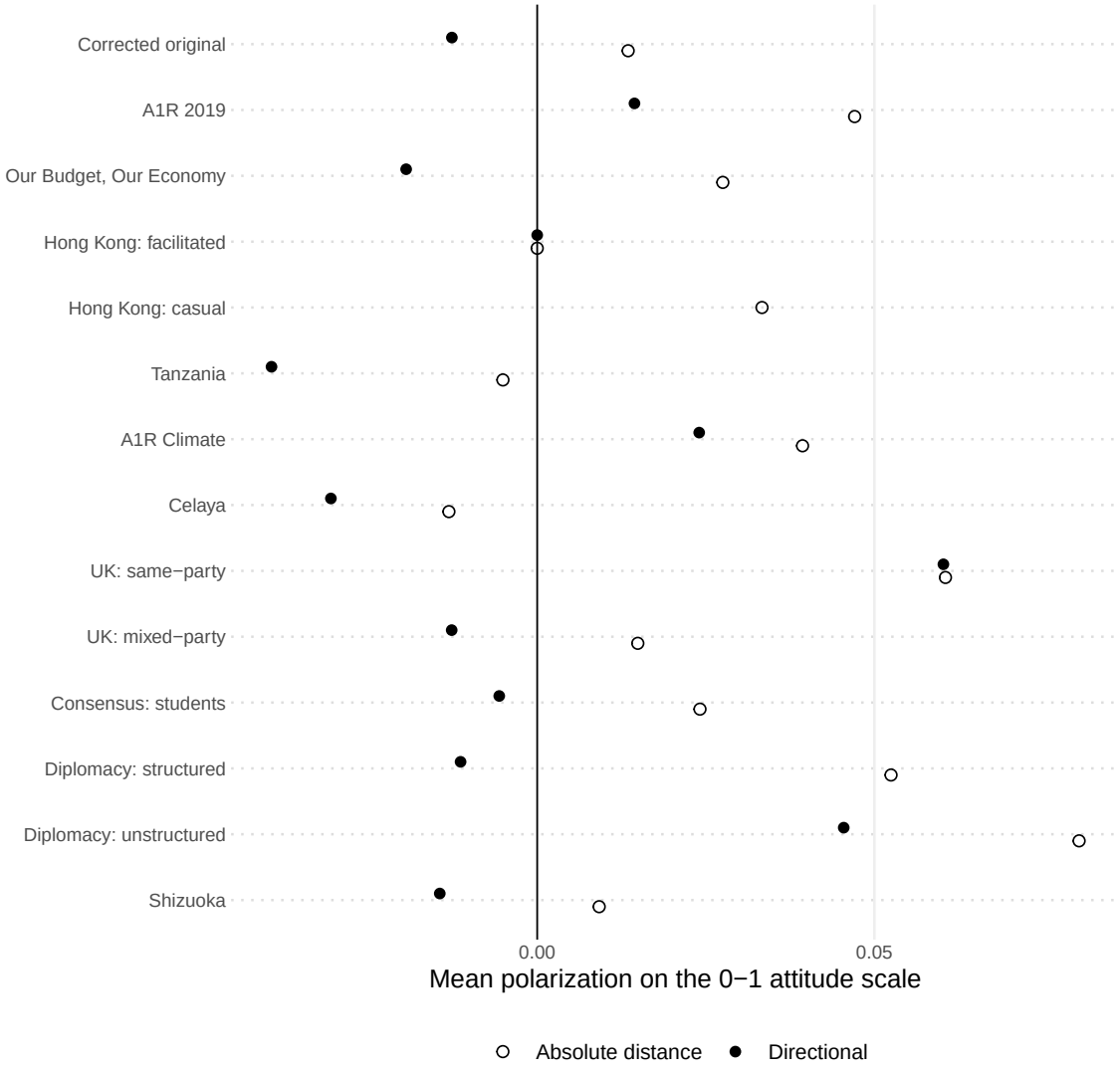


Figure 3: Two definitions of polarization on paired policy responses. Each point is a study-condition mean. Directional and absolute-distance estimates can use different eligible cells, as reported in the preceding table. These are descriptive comparisons; zero marks no average movement under each definition.

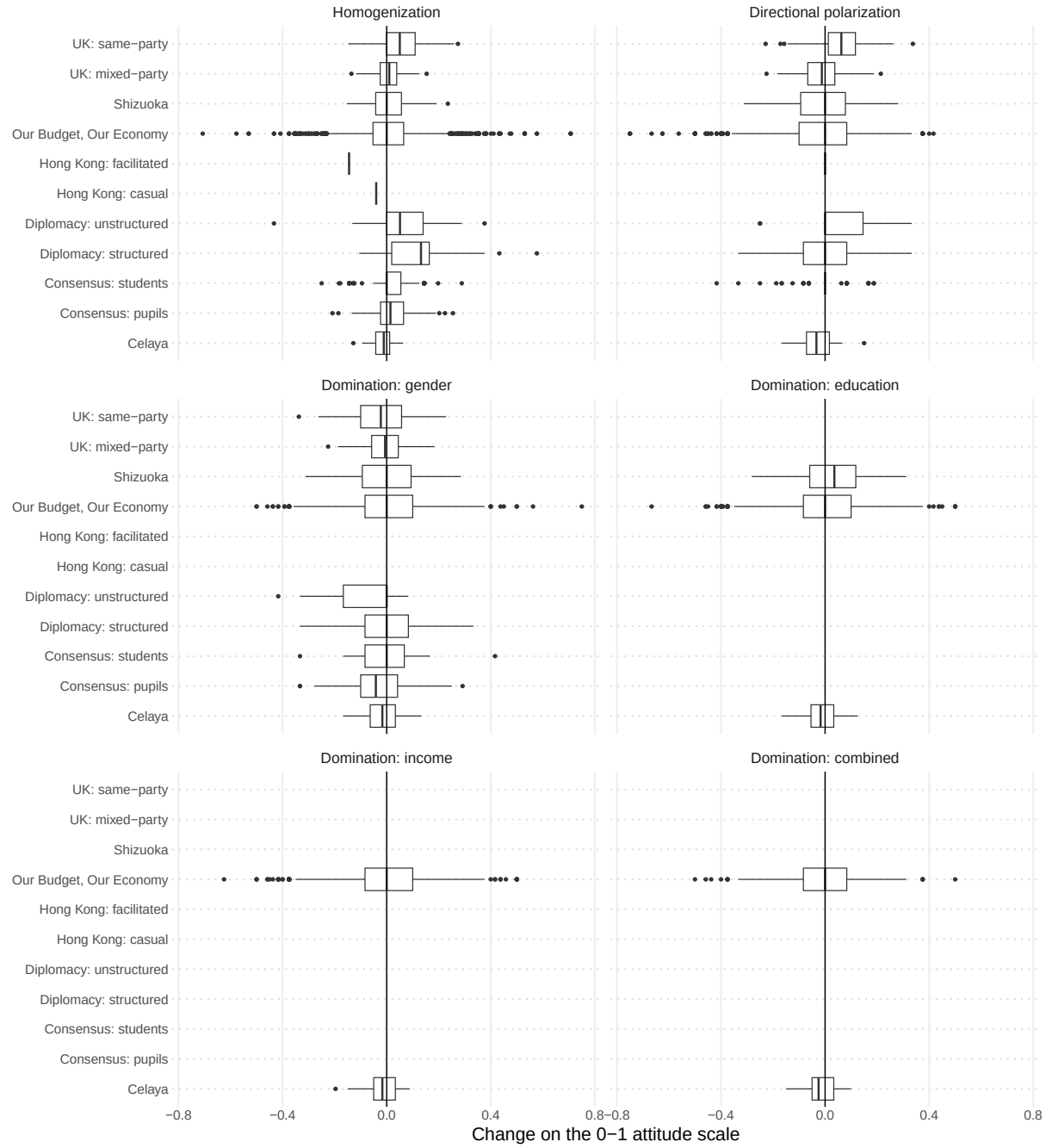


Figure 4: Distributions in the non-DP policy discussions. Boxplots show medians, interquartile ranges, whiskers extending to 1.5 times the interquartile range and observations beyond the whiskers. They summarize eligible paired group-item episodes within each condition; their spread is not uncertainty about the condition mean. Scales are shared across measures.

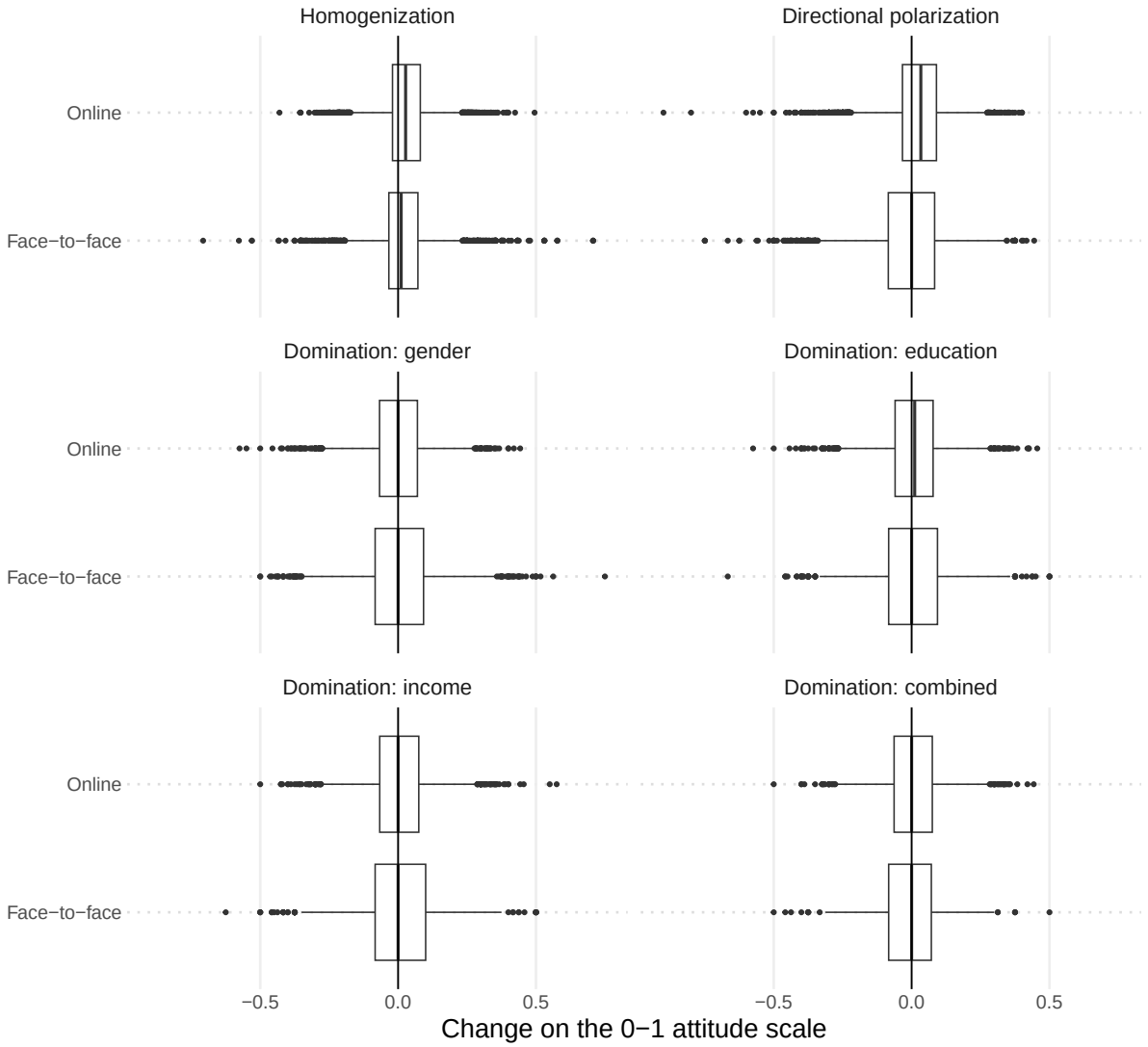


Figure 5: Descriptive mode split, paralleling the original supplement. Paired policy group-item episodes are grouped by the mode of their actual discussion. Mode is confounded with study, topics, group size and participant selection. Boxplots are episode distributions, not causal mode comparisons or confidence intervals.

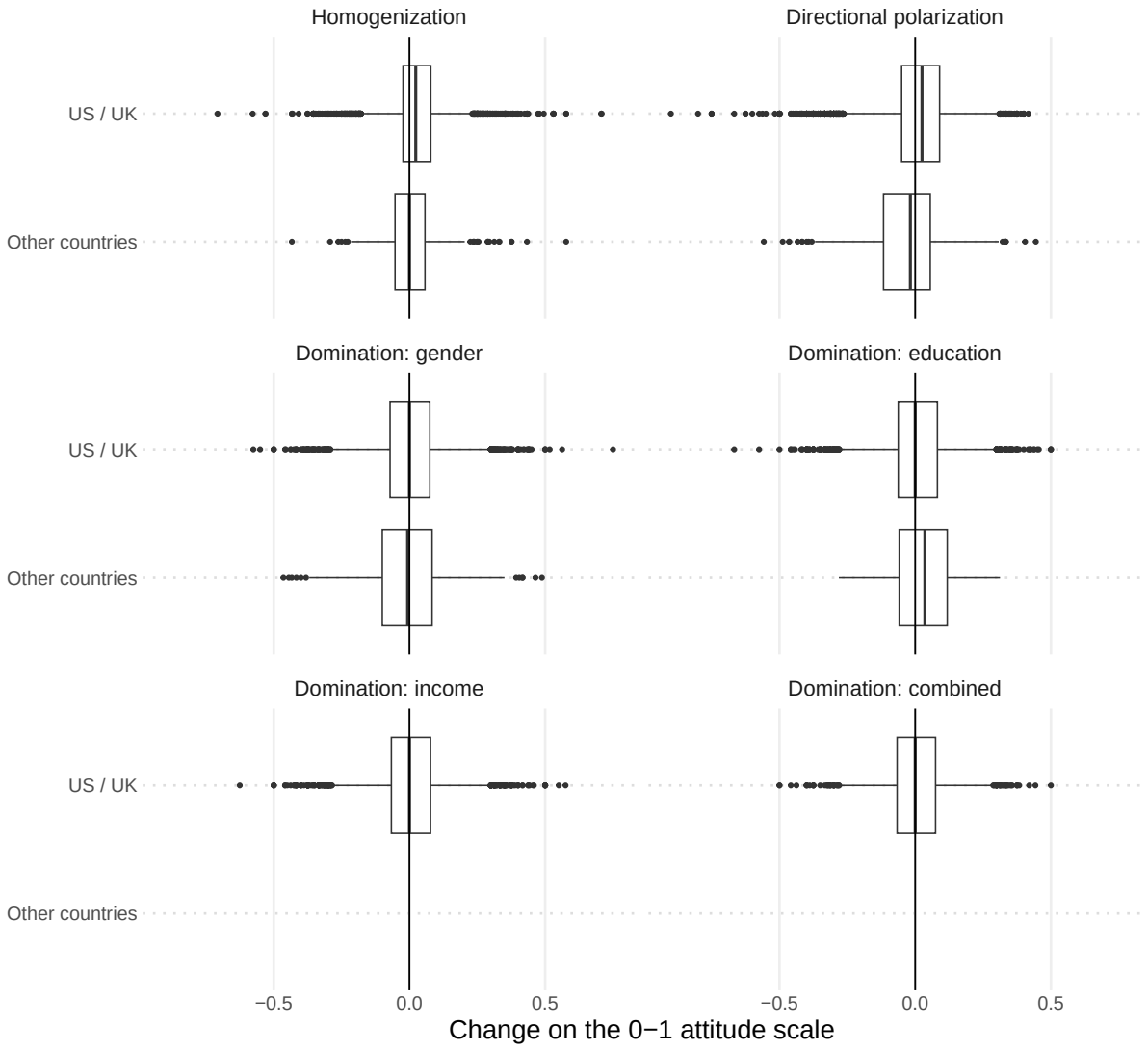


Figure 6: Descriptive country split, paralleling the original supplement. Paired policy group-item episodes from US/UK studies are compared with the remaining countries. The contributing studies and demographic coverage differ; boxplot spread is not sampling uncertainty and the comparison does not isolate a country effect.

Table 8: Directional and absolute-distance polarization

Study / condition	Directional P	Absolute P	N directional	N absolute
Corrected original	-0.013	0.013	2426	2476
A1R 2019	0.014	0.047	1854	1880
Our Budget, Our Economy	-0.019	0.028	1827	1986
Hong Kong: facilitated	0.000	0.000	1	1
Hong Kong: casual	–	0.033	0	1
Tanzania	-0.039	-0.005	542	550
A1R Climate	0.024	0.039	6726	6825
Celaya	-0.031	-0.013	53	56
UK: same-party	0.060	0.061	188	188
UK: mixed-party	-0.013	0.015	149	160
Consensus: pupils	–	–	0	0
Consensus: students	-0.006	0.024	74	82
Diplomacy: structured	-0.011	0.052	22	27
Diplomacy: unstructured	0.045	0.080	22	28
Shizuoka	-0.014	0.009	78	80

Paired policy estimates. An initial midpoint makes directional P undefined but leaves absolute-distance P defined. Midpoint crossings can produce opposite signs. Counts are eligible group-item episodes and can differ between definitions.

D Other outcome constructs

Affective ratings, beliefs, environmental concern and social or stereotype judgments are displayed separately below. They retain the same movement formulas but do not share a substantive estimand merely because their scales can be rescaled to $[0, 1]$. In particular, affective P is not a partisan or religious affect gap.

Table 9: Affective ratings: paired magnitudes and occurrence

Study / condition	H	P	Gender D	Education D	Income D	Combined D
WhatsApp: gen_HH	0.015 / 33.2	-0.068 / 19.6	–	0.003 / 35.3	0.014 / 36.2	0.000 / 34.0
WhatsApp: gen_HM	0.009 / 32.7	-0.081 / 18.6	–	0.009 / 37.7	0.028 / 39.9	0.041 / 44.3
WhatsApp: igr_HH	0.013 / 33.3	-0.087 / 17.1	–	0.008 / 37.1	-0.000 / 35.2	-0.008 / 37.5
WhatsApp: igr_HM	0.016 / 32.9	-0.086 / 16.6	–	-0.032 / 27.1	0.021 / 39.6	0.014 / 36.5
US cross-party conversation	0.019 / 48.7	-0.015 / 42.8	-0.003 / 42.9	–	–	–

Cells are mean / percentage positive, as in Table 3. WhatsApp gen/igr indicate general political/intergroup prompts; HH/HM indicate Hindu–Hindu/Hindu–Muslim dyads. All four conditions share one study family.

Table 10: Beliefs, environmental concern and social judgments

Beliefs

Study / condition	H	P	Gender D	Education D	Income D	Combined D
Shizuoka	0.021 / 50.0	0.028 / 57.1	-0.049 / 20.0	-0.016 / 37.5	–	–

Climate beliefs

Study / condition	H	P	Gender D	Education D	Income D	Combined D
A1R Climate	0.044 / 73.3	0.045 / 72.8	0.001 / 42.3	0.019 / 55.3	0.010 / 51.3	0.013 / 52.2

Environmental concern

Study / condition	H	P	Gender D	Education D	Income D	Combined D
A1R Climate	0.033 / 68.6	0.034 / 68.7	-0.003 / 46.8	0.020 / 61.1	0.013 / 53.7	0.009 / 51.4

Social judgments

Study / condition	H	P	Gender D	Education D	Income D	Combined D
Diplomacy: structured	0.060 / 50.0	-0.000 / 40.0	-0.050 / 20.0	–	–	–
Diplomacy: unstructured	0.037 / 42.9	0.000 / 25.0	0.028 / 33.3	–	–	–

Stereotype judgments

Study / condition	H	P	Gender D	Education D	Income D	Combined D
WhatsApp: gen_HH	-0.001 / 29.9	–	–	-0.013 / 32.3	-0.001 / 33.2	-0.013 / 29.4
WhatsApp: gen_HM	0.026 / 33.8	–	–	-0.009 / 37.2	0.019 / 35.5	0.008 / 36.4
WhatsApp: igr_HH	-0.002 / 29.6	–	–	-0.025 / 26.8	-0.021 / 26.8	0.031 / 33.3
WhatsApp: igr_HM	0.011 / 30.8	–	–	-0.027 / 26.2	0.001 / 31.5	0.011 / 38.3

Cells are mean / percentage positive; constructs are not pooled.

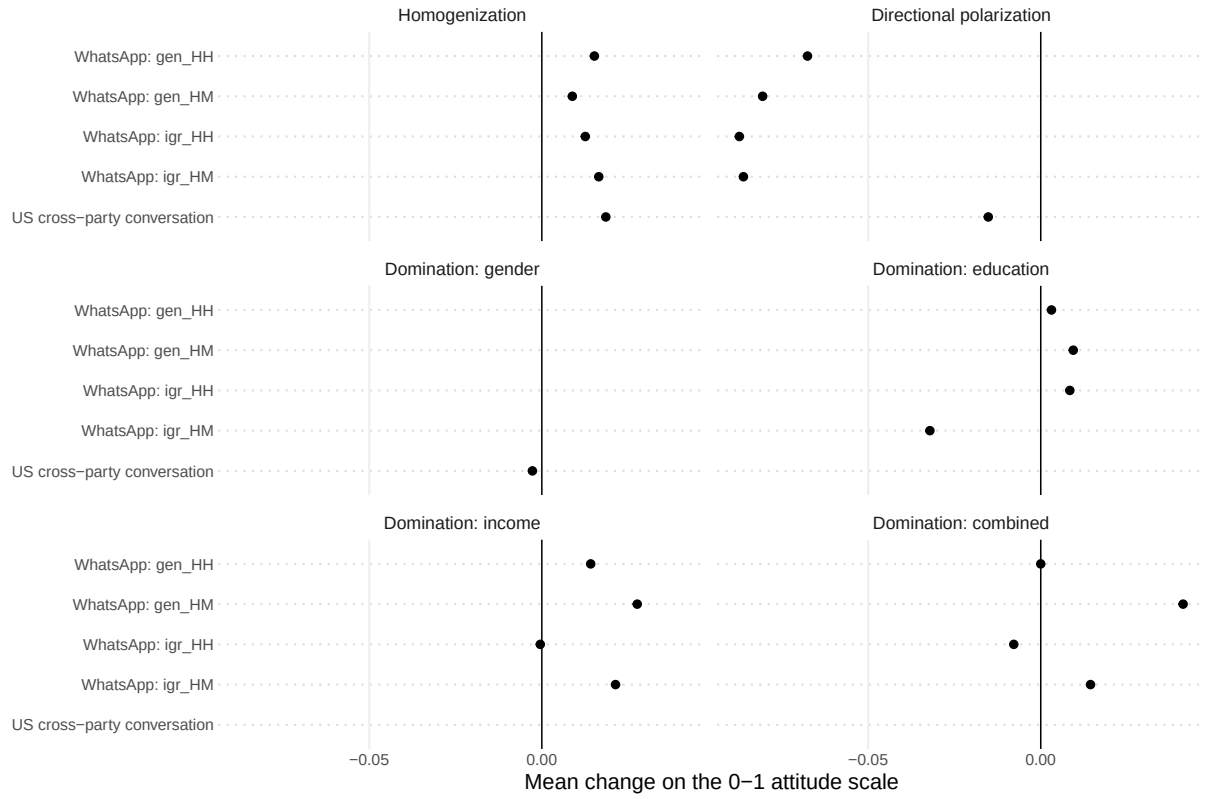


Figure 7: Affective-rating means by observed discussion condition. Points are descriptive paired group-item means, not effects on an in-party/out-party affect gap. Missing panels have no eligible demographic comparison.

E Exploratory comparisons within studies

Exposure to opposing views is one possible explanation to examine, rather than assuming that the DP label itself determines outcomes. The UK experiment varies partisan composition and can inform that hypothesis. Diplomacy varies discussion structure among participants already selected for disagreement; it cannot isolate the effect of exposure. Hong Kong varies facilitation and information in only two groups. The all-study comparison does not identify these mechanisms. We therefore treat the following contrasts as exploratory. A correlation between initial dispersion and subsequent H would also be partly mechanical because initial dispersion is a component of H; it would not establish an exposure effect.

Table 11 puts two comparisons on a 0–100 scale. In the UK study, mixed-party groups exhibit much less directional and absolute polarization than same-party groups. Their mean gender D is also closer to zero, although both signs are negative. Same-party groups homogenize substantially more. Thus less polarization and more agreement do not rank the conditions in the same order. In the Diplomacy experiment, structured discussions have greater homogenization, less polarization on both definitions, and gender D closer to zero than unstructured discussions. These are favorable descriptive patterns if lower extremity and less demographic-directional movement are the criteria; neither agreement nor movement toward neutrality is itself a measure of decision quality.

Table 11: Discussion conditions compared on a 0–100 attitude scale

Study / condition	H	Directional P	Absolute P	Gender D
Corrected original	1.51	-1.27	1.35	0.12
UK: same-party	5.21	6.02	6.05	-1.93
UK: mixed-party	0.52	-1.27	1.49	-0.60
Diplomacy: structured	12.88	-1.14	5.25	1.47
Diplomacy: unstructured	6.41	4.55	8.04	-6.86

Paired policy ratings; equal weights for eligible group-item episodes. Positive H means lower within-group dispersion. Positive P means greater extremity under the indicated definition. Positive gender D means movement along the initial disadvantaged-to-advantaged direction; it is not a direct measure of influence. The UK conditions comprise 345 people in 47 same-party groups and 308 in 40 mixed-party groups. The same 105 people in 35 Diplomacy triads experience both conditions on different topics. Eligible counts vary by measure; gender D uses 162/156 group-items in the UK and 17/17 in Diplomacy. CSV output supplies all counts. The original uses policy indices, whereas the extensions use individual items.

These summaries do not exploit either study's assignment design to estimate a causal contrast. The UK comparison conditions on observed discussion participation and combines the crossed affect-prime treatments; a design-based analysis would also address baseline timing and attrition. Diplomacy requires within-triad contrasts accounting for topic, randomized order and possible carryover. Its topics were selected for disagreement, and demographic comparisons are limited to gender. The current estimates therefore support a description of these samples, not a general ranking of deliberative formats or a test that DPs outperform them.